

Space Integrated Modeling, Simulation, & Analysis (SIMSA) Follow-on Request for Information (RFI)

1.0 Purpose

The purpose of this RFI is to assess options to provide the Modeling, Simulation, Prototyping, & Analysis (MSP&A) work at multiple security classification levels, on multiple security classification environments, at both contractor and customer-designated facilities in support of the United States Space Force (USSF) and related Government entities. To be considered for this work, the contractor must be capable of providing an MSP&A integration / innovation lab, supporting approximately 60 personnel, in the Colorado Springs, CO area that is approved (or will be within the current government fiscal year) to operate up to TS/SCI and SAP levels. The capabilities at this facility should allow multiple vendors to collaborate on analytical objectives supporting the customer(s).

2.0 Background

The Space Warfighting Analysis Center (SWAC) and the Space Security and Defense Program (SSDP) provide force design recommendations to the USSF via analytical studies, leveraging multiple vendors, government civilians, and military personnel.

The analysis conducted at the SWAC, SSDP, and other partner organizations informs strategic decisions on force structure and employment and are based on MSP&A utilizing models of varied fidelity. For the purposes of this RFI, the government is seeking solutions that primarily live in the mission-level MSP&A analytical area.

Responses to this RFI should describe capabilities which support the objectives described in the following sections.

3.0 Requirements

The Government will use this RFI to inform an Acquisition Strategy. This strategy may include one or more possible future Request for Proposals. Specifically, the Government will assess the appropriate alignment of responsibilities between industry partners and the Government. The Government requires:

3.1 Objective 1 – Program Management

The Contractor will provide program management responsible for leading and executing the effort including quality, staffing, and financial and subcontractor management (if necessary). The program manager (PM) will have the authority to make decisions regarding contract work, schedule, and accommodating Government direction. This may include

additional program / project support staff as required to ensure delivery of contract, task order, and associated corporate / United States Government requirements.

3.2 Objective 2 – MSP&A Software Solution

The Contractor will leverage their MSP&A software solution to enable responsive digital engineering (DE); mission engineering (ME); force design and/or solutions development; wargame and exercise planning and execution; training; operational assessment; and modeling, simulation, prototyping & analysis (MSP&A) in support of government objectives. Activities will include product management, development, deployment, security compliance, and maintenance / modernization as required to ensure ongoing relevance and ease-of-use, and to maximize cross-customer integration and utility.

3.3 Objective 3 – Systems Analysis & Design

The Contractor will provide Digital Engineering, Mission Engineering, and/or MSP&A support and/or software development with emphasis to discover, analyze, and assess and/or defend and validate designs and design trades to inform customer force designs, acquisition and/or employment recommendations, architectural trades, and system designs and/or employment. These activities can support Science & Technology (S&T) / Research & Development (R&D), capability/mission area force design analyses, requirements definition, program/system acquisition, and other related activities.

The Contractor may be required to provide any combination of reports, papers, presentations, simulations, models, and data packages, hardware, software, etc. Artifacts may also include optimization routines, algorithms, machine learning (ML), and artificial intelligence (AI) programs, software tools, and user manuals. Artifacts will be delivered or provided to the Government in the format and timeline specified in requirements and may require Government approval as specified in the task order.

Contractor will be required to participate in model verification procedures and at the government's request provide supporting source code, mathematical representations/algorithms, use case results, and other artifacts for third party verification and validation. Contractor will be required to provide and support the execute of a government model verification plan for agents, entities, or systems, instantiated inside the software solution.

3.4 Objective 4 – Systems Operations & Sustainment

The Contractor will provide Digital Engineering, Mission Engineering, and/or MSP&A support and/or software development to discover, analyze, and validate customer systems operations and sustainment. These activities can include systems concepts, TTP

development, analysis, training, education, exercise, and wargaming support, test support, domain or multi-domain operations analyses and/or planning. These activities may also involve support to operationalize, train, transition and/or sustain prototype analysis and/or planning tools.

The Contractor may be required to provide any combination of reports, papers, presentations, simulations, models, and data packages, hardware, software, etc. Artifacts may also include optimization routines, algorithms, machine learning (ML), and artificial intelligence (AI) programs, software tools, and user manuals. Artifacts will be delivered or provided to the Government in the format and timeline specified in requirements and may require Government approval as specified in the task order.

3.5 Objective 5 – Systems / Technical Assessment Integration

The Contractor will apply Digital Engineering, Mission Engineering, Systems Engineering, and/or MSP&A principles as either stand-alone support or to accompany DE, ME, SE, and/or MSP&A activities described in Objectives 2 and/or 3. These technical assessment and/or integration activities involve supporting or managing systems DE, ME, SE, and/or MSP&A definition and/or integration across teams and/or efforts in support of Government objectives. These activities may also involve unbiased analyses and/or systems engineering support to assess tools/environments and provide recommendations, define requirements, or operationalize, train, transition and/or sustain prototyped analysis and/or planning tools.

The Contractor may be required to provide any combination of reports, papers, presentations, simulations, models, and data packages, hardware, software, etc. Artifacts may also include optimization routines, algorithms, machine learning (ML), and artificial intelligence (AI) programs, software tools, and user manuals. Artifacts will be delivered or provided to the Government in the format and timeline specified in requirements and may require Government approval as specified in the task order.

3.6 Objective 6 – Analytical Infrastructure

The Contractor will provide an analytical infrastructure to support OBJECTIVES 2 – 5. This analytical infrastructure may involve Government and/or analytical partner co-use agreements.

4.0 Information Requested

The Space Warfighting Analysis Center (SWAC) conducts critical analytics that directly support the Congressionally mandated authority of the Chief of Space Operations (CSO) as the Chief Architect for the Department of War. To fulfill this mission, SWAC relies on

advanced tools capable of performing rapid, large-scale parametric analyses to evaluate how space-based capabilities contribute to the Joint fight. This requires sophisticated agent-based simulations that enable a deep understanding of Satellite Communications (SATCOM), Navigation Warfare (SATNAV), information data flows, and other mission areas, all aligned with authoritative models.

SWAC will require an analytic tool(s) that is designed to meet these demands, and provides the necessary fidelity to support complex assessments while maintaining the flexibility to adapt quickly to evolving analytic requirements from Space Force, Department of the Air Force (DAF), and Department of War (DoW) leadership. Over the past two and a half years, analytical tool(s) have been integral to SWAC's efforts to conduct high-value functional analyses, identifying where Space Force funding achieves the greatest impact in support of operational objectives. This work has required the development of innovative methodologies to integrate such tool(s) with tool(s) from other organizations within a classified cloud environment, enabling large-scale data analytics involving massive numbers of agents, and the need for such tool(s) that have standardized communication protocols to seamlessly interface with authoritative models and partner tools, such as those from the Office of the Secretary of Defense for Research and Engineering (OSD R&E). A disruptive transition of any kind would have significant negative impacts to SWAC's ability to maintain current operations, as analytics are central to the mission and proven capabilities and established integration make it indispensable for ensuring continuity and effectiveness in supporting the Space Force's critical objectives.

In response to this Request for Information, please respond to the following:

1. Does your company have a Modeling, Simulation, Prototyping, & Analysis (MSP&A) software solution focused on the space domain?

a. If yes, describe the current functionality in terms of MSP&A support performed, and which organizations you have supported, additionally:

1. Are there licensing costs associated with your MSP&A software solution?" (ex: Ansys software used by OTTI is free for CIV/MIL use, contractor support must pay)

2. Are there licensing costs associated with your MSP&A software solution?" (ex: Ansys software used by OTTI is free for CIV/MIL use, contractor support must pay)

3. What is the current method for modifying or adding capability to your software solution?

2. Does your MSP&A software support mission-level space domain analysis?

a. If yes, describe the mission-level analysis carried out, and which organizations you have supported.

3. Describe your company's past performance in supporting USSF force design analyses.

4. Describe your MSP&A software solution's capabilities in modeling space domains such as, but not limited to, Electronic Warfare, NAVWAR, Spectrum Warfare, Space Control, etc.

5. Describe your MSP&A software solution for performing architectural resiliency analyses.

6. Describe your MSP&A software solution for providing kill chain analyses in support of wargaming activities such as the SWAC / SSDP Battle School series.

7. Do you have demonstrable past performance integrating your MSP&A software solution with other tools, particularly Acropolis, RCADE, and SPySE?

a. If yes, describe the past performance.

8. Does your MSP&A tool currently support Kafka message bus services for data exchange?

9. Describe your ability to leverage AI/ML to support technical analysis objectives.

10. Describe your company's ability to provide an integration / innovation lab where the government customers and associated analytical partners (approximately 60 personnel) can collaborate and integrate their analytical software and produce integrated analytical results.

11. Does your facility currently have, or will have, within 6 months, connectivity to a classified cloud environment, such as Microsoft Azure's PLx, and the ability to operate at TS/SCI and SAP levels?